

CSE 420 Compiler Design Quiz 3 (SET B)

Student ID: _____ Name: _____

Section: _____ Date: _____

Consider the SDT below:

$$P \rightarrow \left\{ \begin{array}{l} \text{tab} = \text{new SymbolTable}(); \quad \text{off} = 200; \\ \text{stTab} = \text{new Stack}(); \quad \text{stOff} = \text{new Stack}(); \end{array} \right\} \quad D$$

$$D \rightarrow T \text{ id}; \left\{ \begin{array}{l} \text{tab.insert(id.lexeme, T.t, T.w, off)}; \\ \text{off} = \text{off} + \text{T.w}; \end{array} \right\} \quad D$$

$$D \rightarrow \epsilon$$

$$T \rightarrow B \left\{ \begin{array}{l} C.bt = B.t; \\ C.bw = B.w; \end{array} \right\} \quad C \left\{ \begin{array}{l} T.t = C.t; \\ T.w = C.w; \end{array} \right\}$$

$$T \rightarrow \text{record} \left\{ \begin{array}{l} \text{stTab.push(tab);} \quad \text{stOff.push(off);} \\ \text{tab} = \text{new SymbolTable}(); \quad \text{off} = 20; \end{array} \right\} \quad D \left\{ \begin{array}{l} T.t = \text{record(tab);} \quad T.w = \text{off}; \\ \text{tab} = \text{stTab.pop}; \\ \text{off} = \text{stOff.pop}; \end{array} \right\}$$

$$B \rightarrow \text{int} \left\{ \begin{array}{l} B.t = \text{int}; \\ B.w = 10; \end{array} \right\}$$

$$B \rightarrow \text{float} \left\{ \begin{array}{l} B.t = \text{float}; \\ B.w = 20; \end{array} \right\}$$

$$C \rightarrow [\text{num}] \left\{ \begin{array}{l} C_1.bt = C.bt; \\ C_1.bw = C.bw \end{array} \right\} \quad C_1 \left\{ \begin{array}{l} C.t = \text{array}(\text{num.value}, C_1.t); \\ C.w = \text{num.value} * C_1.w \end{array} \right\}$$

$$C \rightarrow \epsilon \left\{ \begin{array}{l} C.t = C.bt; \\ C.w = C.bw; \end{array} \right\}$$

Q1) Draw Annotated Parse Tree for this input:

[10 Marks]

```
record {
    float [3] data;
    record {
        int x;
    } p2;
} r2;
```

Q2) Write Three Address Code for this input code:

[5 Marks]

```
y = func( func(arr[j + 1], i) , arr[i * 2])
```

Here `arr` is a float array, and `func` is a function.

Answer: